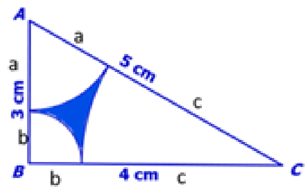
Term 2 Week 2



1. $a + c = 5$
   $a + b = 3$
   $b + c = 4$
   Solving, gives $a = 2, b = 1, c = 3$
   Area of the triangle: $A = \frac{1}{2} \times 3 \times 4 = 6cm^2$

   Use trigonometry to find the angles at each A, then since B = 90 we can find C. Then use these to calculate the area of each sector.

   $\sin A = \frac{4}{5}$
   $A = \sin^{-1}\frac{4}{5} = 53.13$
   $B = 90 - 53.13 = 36.87$
   Area of arc A = $\frac{53.13}{360} \times \pi(2)^2 = 1.85$
   Area of arc B = $\frac{90}{360} \times \pi(1)^2 = 0.79$
   Area of arc C = $\frac{36.87}{360} \times \pi(3)^2 = 2.90$
   Shaded area = $6 - 1.85 - 0.79 - 2.90 = 0.46$

2.
$$\frac{dV}{dt} = kA \qquad \frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} \qquad V = \frac{4}{3}\pi r^3 \qquad A = \pi r^2$$

   Combining:
   $kA = 4\pi r^2 \times \frac{dr}{dt}$
   Therefore, $\frac{dr}{dt} = k$

   Separating variables and integrating:
   $\int dr = \int k\, dt$
   $r = kt + c$

   Finding initial radius and radius at time m:

$$t = 0, V = 1 \qquad\qquad t = m, V = \frac{1}{2}$$

$$1 = \frac{4}{3}\pi r^3 \qquad\qquad \frac{1}{2} = \frac{4}{3}\pi r_m^3$$

$$r = \sqrt[3]{\frac{3}{4\pi}} \qquad\qquad r_m = \sqrt[3]{\frac{3}{8\pi}} = 0.49$$

Now we can find k and c:
When $t = 0$:
$0.62 = k \times 0 + c$
$c = 0.62$
$r = kt + 0.62$

When $t = m$:
$0.49 = km + 0.62$
$km = -0.13$
$k = -\frac{0.13}{m}$

Therefore $r(t) = -\frac{0.13}{m}t + 0.62$

When the sphere has dissolved, the radius equals zero.

$0 = -\frac{0.13}{m}t + 0.62$
$\frac{0.13}{m}t = 0.62$
$t = \frac{0.62m}{0.13} = 4.77m$

3. $f(n+2) = 3^{n+3} + 2^{n+2}$
Which we can rewrite as:
$3^2 \times 3^{n+1} + 2^2 \times 2^n = 9 \times 3^{n+1} + 4 \times 2^n$

$f(n+1) = 3^{n+2} + 2^{n+1}$
Which we can rewrite as:
$3 \times 3^{n+1} + 2 \times 2^n$

Forming an equation:
$9 \times 3^{n+1} + 4 \times 2^n = a(3 \times 3^{n+1} + 2 \times 2^n) + b(3^{n+1} + 2^n)$
$3 \times 3^{n+1} + 2 \times 2^n = 3a \times 3^{n+1} + 2a \times 2^n + b \times 3^{n+1} + b \times 2^n$

Factorising:
$9 \times 3^{n+1} + 4 \times 2^n = (3a+b)3^{n+1} + (2a+b)2^n$

Therefore, we know:
$3a + b = 9$
$2a + b = 4$

Solving simultaneously, we get $a = 5, b = -6$

4. Use partial fractions to rewrite fraction:
$\frac{1}{4k^2-1} = \frac{1}{(2k-1)(2k+1)} = \frac{A}{2k-1} + \frac{B}{2k+1}$

   Multiplying through by $(2k-1)(2k+1)$, we get:
$1 = A(2k+1) + B(2k-1)$
$1 = 2Ak + A + 2Bk - B$
$2a + 2b = 0 \Rightarrow a + b = 0$
$A - B = 1$
$2A = 1$
$A = \frac{1}{2}, B = -\frac{1}{2}$

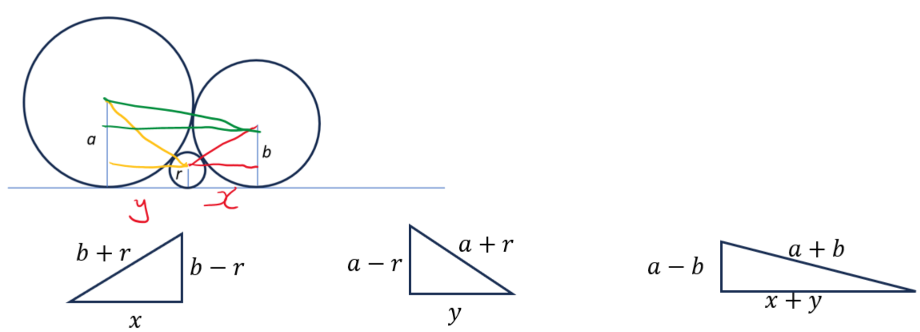
   So, our sum can be written as:

$$\sum_{k=1}^{\infty} \frac{1}{2(2k-1)} - \frac{1}{2(2k+1)}$$

   Substitute a few values of k to see what the series looks like:
$(\frac{1}{2} - \frac{1}{6}) + (\frac{1}{6} - \frac{1}{10}) + (\frac{1}{10} - \frac{1}{14}) + ...$

   We can see that all the terms after the $\frac{1}{2}$ will cancel out. This means the sum to infinity is $\frac{1}{2}$.

5. Adding lines to create a series of right-angle triangles:



Form expressions for x and y:
$x^2 = (b + r)^2 - (b - r)^2$
$x^2 = b^2 + 2br + r^2 - b^2 + 2br - r^2$
$x^2 = 4br$
$x = 2\sqrt{br}$
$y^2 = 4ar$
$y = 2\sqrt{ar}$

$(a + b)^2 = (a - b)^2 + (x + y)^2$
$(a + b)^2 = (a - b)^2 + (2\sqrt{br} + 2\sqrt{ar})^2$
$a^2 + b^2 + 2ab = a^2 - 2ab + b^2 + 4br + 4ar + 8\sqrt{abr^2}$
$4ab = 4br + 4ar + 8\sqrt{abr^2}$
$ab = br + ar + 2\sqrt{abr^2}$
$ab = br + ar + 2r\sqrt{ab}$
$ab = r(a + b + 2\sqrt{ab}$
$ab = r(\sqrt{a} + \sqrt{b})^2$

Rearranging:
$\frac{1}{r} = \frac{(\sqrt{a}+\sqrt{b})^2}{ab}$
$\frac{1}{\sqrt{r}} = \frac{\sqrt{a}+\sqrt{b}}{\sqrt{ab}}$
$\frac{1}{\sqrt{r}} = \frac{\sqrt{a}}{\sqrt{a}\sqrt{b}} + \frac{\sqrt{b}}{\sqrt{a}\sqrt{b}}$
$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}$
As required.

6. Set

$$x = \sqrt[3]{2 + \sqrt{5}} + \sqrt[3]{2 - \sqrt{5}}$$

Cube both sides

$$x^3 = 2 + \sqrt{5} + 3\sqrt[3]{(2 + \sqrt{5})^2}\sqrt[3]{2 - \sqrt{5}} + 3\sqrt[3]{2 + \sqrt{5}}\sqrt[3]{(2 - \sqrt{5})^2} + 2 - \sqrt{5}$$

$$x^3 = 4 + 3\sqrt[3]{(2 + \sqrt{5})(2 + \sqrt{5})(2 - \sqrt{5})} + 3\sqrt[3]{(2 + \sqrt{5})(2 - \sqrt{5})(2 - \sqrt{5})}$$

Since $(2 + \sqrt{5})(2 - \sqrt{5}) = -1$, we can factor out -3, giving us:

$$x^3 = 4 - 3(\sqrt[3]{2 + \sqrt{5}} + \sqrt[3]{2 - \sqrt{5}})$$

What is left in the brackets is our original expression, so we can rewrite as:

$$x^3 = 4 - 3x$$

Solving the cubic $x^3 + 3x - 4 = 0$ we get $x = 1$, the simplified value of our original expression.